

Computational creativity

Given the difficulty of defining creativity and employing any rough definitions into the field of computers, artificial intelligence experts sought to come up with a new field of science to further develop this conundrum. It was called computational creativity and at the forefront were the same three gentlemen who pioneered the usage of reasoning and the 'neat' approach to AI: Robert Simon, Alan Newell and J. C. Shaw.

For the sake of computing and artificial intelligence, they decided to focus on the two aspects of creativity that would be most important in an intelligent agent: novelty and usefulness. This disregards the field of creativity for the sake of art or for no real objective at all, and we'll come to that later.

When an AI is posed a problem, Simon, Newell and Shaw came up with four guidelines to define computational creativity:

- i) The answer is novel and useful (either for the individual or for society)
- ii) The answer demands that we reject ideas we had previously accepted
- iii) The answer results from intense motivation and persistence
- iv) The answer comes from clarifying a problem that was originally vague

How does this help the AI? Well, much like Asimov's three laws of robotics, these guidelines make sure that the AI will look for creativity. The first guideline (novel and useful) clarifies the objective of the solution being sought. The second guideline (rejecting previously successful ideas) insists that the AI abandon any thought process that it has already gone through to come to a successful conclusion, thus ensuring that the solution will be new. The third guideline instructs the AI to keep looking for a solution even if all earlier paths have been tried and failed, since the machine would otherwise just report a negative result. And the fourth guideline, perhaps the most intriguing, tells the AI that the problem as stated may not have completely meant as stated – i.e. make the machine re-examine the true nature of the problem and not just its semantic meaning, thus opening its horizons for different interpretations and problems.

Mimicking human creativity for problem solving

Given the parameters that Simon, Newell and Shaw laid down, research in applying creativity to computers began in earnest. There have been several programs over the years which have successfully applied this to problem-solving, but none has been as interesting as the recent work at Rensselaer Polytechnic Institute in the US.